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DANG: Prototype-Anchored Foundation Features for Label-Efficient and Robust Polyp Segmentation

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ABSTRACT Polyp segmentation networks that reach about 93% mean Dice on in-domain colonoscopy images degrade when only a few frames are labelled, on tiny or low-contrast lesions, and under changes in acquisition conditions. We present DANG, a segmentation network that pairs a trainable ConvNeXt-Tiny encoder with a frozen DINOv2-Base foundation model and uses the foundation features through three components: a dual-prototype margin head, a prototype-anchored spatial support envelope that suppresses false positives far from the lesion, and an uncertainty-guided strip decoder. We train on 40 or 800 Kvasir-SEG images, test on deduplicated external and stress cohorts over five paired random seeds, and retrain PraNet, Polyp-PVT and EMCAD under the same protocol. With 40 labelled images DANG leads every retrained model on all external evaluations, by at least 7.1 percentage points of mean Dice and 4.8 points of boundary F1, and most of these differences remain significant after Holm correction. A control that adds the same frozen DINOv2 by feature concatenation stays at baseline level, so the frozen features alone do not explain the gain. With 800 labelled images DANG is statistically tied with Polyp-PVT and EMCAD. On the standard 1,450-image training split it matches a faithful Polyp-PVT reproduction on four test sets and exceeds it on ETIS-LaribPolypDB (84.4% versus 78.9% mean Dice; three seeds, uncorrected). These gains cost inference speed: DANG runs at 17 frames per second on an NVIDIA T4, two to two and a half times slower than the retrained models.

INDEX TERMS Colonoscopy, deep learning, foundation models, label-efficient learning, medical image segmentation, polyp segmentation, prototype learning, robustness.

I. INTRODUCTION

COLORECTAL cancer is the third most common cancer and the second leading cause of cancer death worldwide [1], and removing adenomatous polyps during colonoscopy lowers colorectal-cancer mortality [2]. Pixel-level polyp segmentation supports this by delineating lesion margins. On the in-domain benchmark Kvasir-SEG [3], however, a plain ConvNeXt-Tiny [4] with a Feature Pyramid Network [5] already reaches about 93% mDice, and in our experiments the proposed model and this baseline differ by 0.03 pp in domain (Section V-C). In-domain scores therefore say little about robustness.

Deployment stresses a model along three axes that in-domain benchmarks do not exercise:

1) **Scarce labels.** Pixel-accurate contours require expert time, so a new centre may label only tens of frames. Trained on 40 images, CNNs overfit to source textures and their boundaries degrade on external centres.
2) **Tiny and camouflaged lesions.** Flat or diminutive polyps covering under 1.5% of the frame, or with low contrast to the surrounding mucosa, are fragmented or missed by stride-16/32 features, especially on ETIS-LaribPolypDB [6] and CVC-ColonDB [7].
3) **Acquisition shift.** Procedures switch between white-light and NBI-like spectra and suffer illumination fall-off, sensor noise, specular glare and motion blur.

Vision foundation models such as DINOv2 [8] and SAM [9] provide features that transfer across domains, but their 14- or 16-pixel patch grids are too coarse for lesion boundaries. Concatenating such features with a CNN is the obvious remedy; we show that it does not help when labels are scarce. *We therefore do not target in-domain state of the*

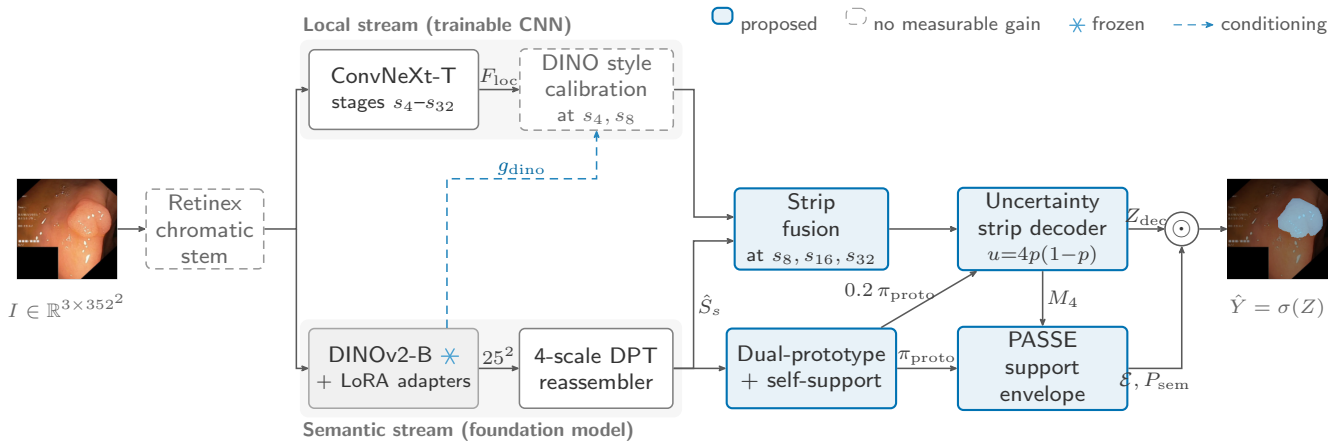


FIGURE 1. **DANG fuses a trainable CNN stream with a frozen foundation-model stream and gates the output with a prototype-derived support envelope. A 352 × 352 frame splits into a ConvNeXt-Tiny local stream and a frozen DINOv2-Base semantic stream with low-rank adapters. Tokens from four depths are reassembled at four scales; strip fusion aligns the two streams at strides 8–32; the dual-prototype head produces the margin map π_{proto}, which enters the decoder output Z_{dec} with weight 0.2; PASSE combines π_{proto} with the coarsest decoder logit M_4 into an envelope $\mathcal{E}$ and core map P_{sem} that modulate Z_{dec} (Eq. 6). Blue boxes are the proposed components; the dashed Retinex stem and style calibration are part of the evaluated model but show no measurable gain (Section V-C). Thumbnails: a Kvasir test frame and its ground-truth mask, for illustration only.**

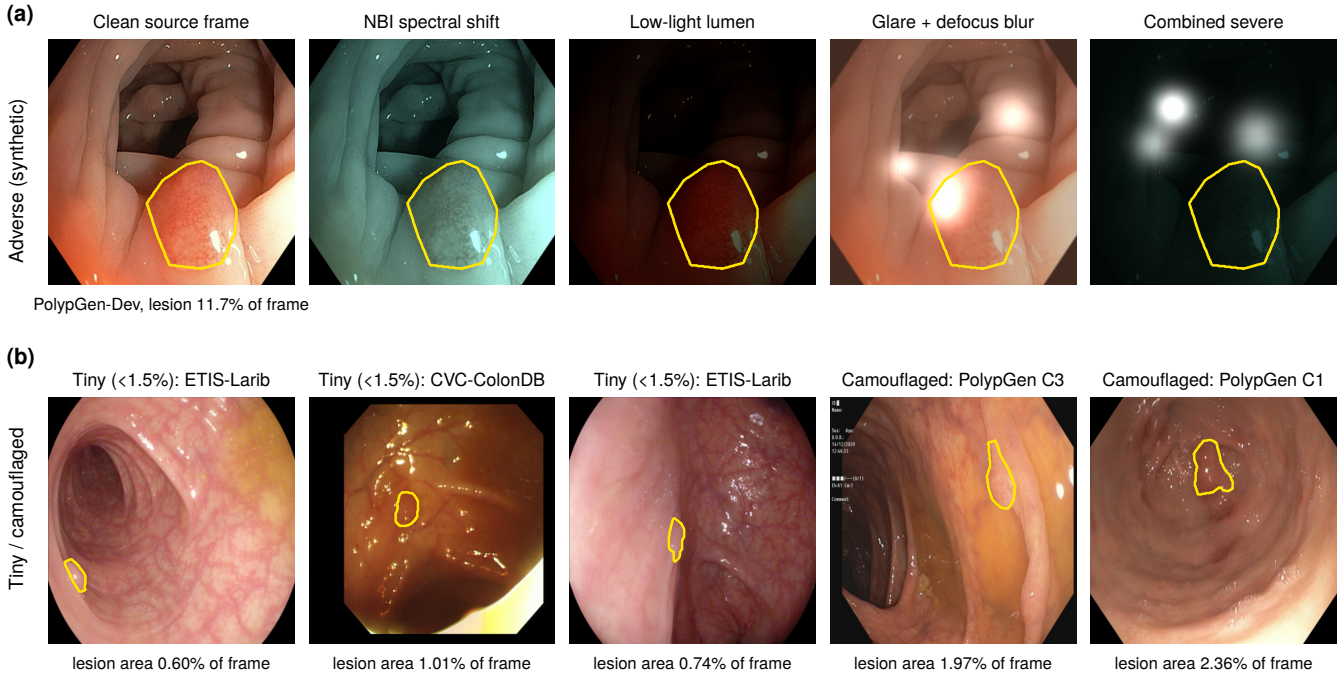


FIGURE 2. **The two stress cohorts. (a) One PolypGen-Dev frame, clean and under the four synthetic corruptions of the adverse cohort; in the benchmark each frame receives one of the four. (b) Five tiny or camouflaged lesions drawn at random (three tiny, two camouflaged), not curated. Yellow contours are ground truth. Predictions on such frames are shown in Figure 3.**

art; we ask how frozen foundation features should be used when only a few frames are labelled.

We present **DANG** (Figure 1) and make three contributions:

- **A decoder that uses foundation features through prototypes.** Multi-depth DINOv2 tokens feed a dual-prototype margin head with test-time self-support; the **prototype-anchored spatial support envelope (PASSE)** turns this margin into a mask that suppresses false positives far from the lesion and reinforces low-contrast lesion cores; and an uncertainty-guided strip decoder refines boundaries where the coarse prediction is ambiguous.
- **A controlled evaluation protocol.** A deduplicated multi-centre benchmark with a 40-label training regime and two stress cohorts (Figure 2); PraNet [10], Polyp-PVT [11] and EMCAD [12] retrained under the same data, schedule and evaluation; five paired seeds with

Holm-corrected tests; and a standard-split experiment that links the protocol to published numbers.

- **Findings, positive and negative.** A same-size concatenation control shows that the frozen features alone do not explain DANG's gain. DANG leads all retrained models with 40 labels, ties the strongest of them with 800 labels, and costs 2–2.5× their inference time; its Retinex stem shows no measurable gain.

II. RELATED WORK

a: Polyp segmentation.

Architectures progressed from U-Net [13] and UNet++ [14] to reverse-attention networks: PraNet [10] mines boundary cues with parallel partial decoders and reverse attention, and CaraNet [15] adds axial context for small objects. Transformer encoders improved multi-centre generalisation in Polyp-PVT [11] and TransFuse [16], and EMCAD [12] pairs a PVTv2 encoder with an efficient multi-scale convolutional attention decoder. Label-efficient work on polyps has mostly been semi-supervised; FPGM [17], for example, uses a frequency prior on polyp edges to augment unlabelled images and improve generalisation to new centres. DANG instead uses no unlabelled data and draws its prior from a frozen foundation model.

b: Foundation models for polyps.

SAM applied without prompts segments polyps poorly [18], and MedSAM [19] fine-tunes SAM on a large medical corpus but requires box prompts. SAM-Mamba [20] and SAM-MaGuP [21] adapt SAM with Mamba-based priors for zero-shot polyp generalisation, at about 103M parameters and 423 GFLOPs [22]. A second line distils foundation models into small networks: Polyp-DiFoM [23] distils SAM and DINOv2 into lightweight baselines, LiteBounD [22] adds boundary-guided frequency-aware distillation, and Lite-π [24] aligns foundation-model-specific prototypes with a lightweight network for real-time inference. DANG keeps the foundation model at inference and studies how to use its features when labels are scarce; distillation is complementary and could compress it.

c: Prototype-based segmentation.

Few-shot semantic segmentation classifies query pixels by their similarity to class prototypes computed from support images [25]. Self-support prototypes [26] are built from confident features of the query image itself, which narrows the gap between support and query appearance. DANG adopts this self-support idea inside a fully supervised segmenter: learned prototypes and ground-truth anchors give a prior, and image-specific self-support centroids adapt it at test time.

d: Adaptation and directional operators.

Low-rank adaptation (LoRA [27]), its convolutional variant for SAM (Conv-LoRA [28]) and dense-prediction heads (DPT [29]) adapt frozen ViTs to segmentation; DANG uses convolutional low-rank adapters and a DPT-style reassembler. State-space models (Mamba [30], VMamba [31], U-Mamba [32]) capture long-range context with directional scans; our strip operator approximates directional context with depthwise strip convolutions.

e: Robustness evaluation.

Our synthetic adverse cohort follows the corruption-benchmark design of ImageNet-C [33]: deterministic, parametrised corruptions applied only at test time.

III. METHOD

DANG maps an RGB frame $I \in \mathbb{R}^{3\times352\times352}$ to a polyp probability map $\hat{Y} = \sigma(Z)$ (Figure 1). We write $X^{\uparrow n}$ for bilinear upsampling of a map X to $n \times n$.

A. TWO STREAMS

The local stream is ConvNeXt-Tiny, producing features $F^{(s)}$ at strides $s \in \{4, 8, 16, 32\}$. The semantic stream is DINOv2-Base, frozen, applied to the frame resized to 350×350 so that its 14-pixel patches form a 25×25 grid. Rank-4 convolutional low-rank adapters [28] on its last four blocks are trained. The evaluated model also contains a Retinex chromatic stem [34], which removes specular highlights and divides out a smooth illumination estimate before both streams, and a style calibration of $F^{(4)}$ and $F^{(8)}$ conditioned on the global DINOv2 vector (with MixStyle [35]). Neither is trained on corrupted images, and Section V-C shows that together they give no measurable gain.

B. MULTI-SCALE REASSEMBLY

We take DINOv2 tokens from blocks $\{2, 5, 8, 11\}$ and project each to a 192-channel map S_ℓ on the 25×25 grid. A learned depth-mixing matrix $W = \mathrm{Softmax}(\Theta) \in \mathbb{R}^{4\times4}$, with Θ learned logits, produces one semantic map per stride,

$$\tilde{S}_s = \sum_\ell W_{s,\ell} S_\ell, \qquad \hat{S}_s = \phi_s\big(\mathrm{Up}_s(\tilde{S}_s, F^{(s)})\big), \tag{1}$$

where ϕ_s is a 3×3 convolution block and Up_s resizes to stride s. At strides 8 and 16, Up_s predicts 3×3 reassembly kernels from $F^{(s)}$, initialised to bilinear interpolation, so that upsampled semantic features align with CNN edges; at the other strides it is bilinear.

C. DUAL-PROTOTYPE HEAD

The head concatenates the three deepest maps, projects them to 128 channels and normalises each position to a unit embedding $e_{u,v}$. It compares $e_{u,v}$ with $K = 8$ learned foreground prototypes p_k^+ and $K = 8$ background prototypes p_k^- (unit vectors, orthogonally initialised), and with a foreground anchor a^+ and background anchor a^-:

$$\pi_{\mathrm{prior}}(u,v) = \tfrac{\tau}{\beta}\Big(\log\textstyle\sum_k e^{\beta\langle e_{u,v}, p_k^+\rangle} - \log\sum_k e^{\beta\langle e_{u,v}, p_k^-\rangle}\Big) + w_{\mathrm{a}}\big(\langle e_{u,v}, a^+\rangle - \langle e_{u,v}, a^-\rangle\big) + b. \tag{2}$$

The log-sum-exp with sharpness β (fixed) is a soft maximum over prototypes; the temperature τ, the bias b and the bounded anchor weight w_a are learned. During training only, $a^{\pm}$ are exponential moving averages of the embeddings at positions where the ground-truth mask, average-pooled to 25×25, is mostly foreground or mostly background. At training and test time the head then forms image-specific self-support centroids [26] weighted by the prior's own confidence, $c^{+} = \mathrm{Norm}(\sum_{u,v} q_{u,v}^2 e_{u,v})$ and $c^{-} = \mathrm{Norm}(\sum_{u,v}(1-q_{u,v})^2 e_{u,v})$ with $q = \sigma(\pi_{\mathrm{prior}})$, and outputs the margin map

$$\pi_{\mathrm{proto}} = \pi_{\mathrm{prior}} + w_{\mathrm{ss}}\big(\langle e, c^{+}\rangle - \langle e, c^{-}\rangle\big), \tag{3}$$

with a learned bounded weight w_{ss}. Constant values are listed in Appendix A.

D. STRIP FUSION AND UNCERTAINTY-GUIDED DECODER

At strides 8–32 the local features $F^{(s)}$ and semantic maps $\hat{S}_s$ are fused by a gated directional operator applied to a feature map X with conditioning map C from the other stream:

$$\begin{aligned}\mathrm{Strip}(X, C) = X &+ g_h \odot \mathcal{K}_{7\times1}\mathcal{K}_{1\times11}(X)\\ &+ g_v \odot \mathcal{K}_{1\times7}\mathcal{K}_{11\times1}(X)\\ &+ g_d \odot \mathcal{K}_{3\times3,d=3}(X).\end{aligned} \tag{4}$$

Here $\mathcal{K}$ are depthwise convolutions (horizontal and vertical strips, and a dilated 3×3 branch for diagonal context) and $(g_h, g_v, g_d) = \sigma(\mathrm{Conv}_{1\times1}([X, C]))$ are gates. The top-down decoder has four stages that output logits $M_4, \ldots, M_1$ at strides $32, \ldots, 4$. Each stage converts the upsampled logit of the coarser stage into a probability p, computes the boundary uncertainty $u = 4p(1-p)$, which peaks on ambiguous borders where $p = 0.5$, and an outer ring $\mathrm{MaxPool}_{5\times5}(p) - p$ just outside the predicted lesion, and uses both maps to gate its strip operator and a Sobel-based edge branch. The stride-4 decoder output is $Z_{\mathrm{dec}} = 0.8\, Z_{\mathrm{head}} + 0.2\, \pi_{\mathrm{proto}}^{\uparrow 88}$, where Z_{head} combines M_1 with a head over all four stages.

E. PROTOTYPE-ANCHORED SPATIAL SUPPORT ENVELOPE (PASSE)

CNN texture far from the lesion can cross the decision threshold and produce isolated false positives. PASSE builds a coarse semantic probability $P_{\mathrm{sem}} = \sigma\big(\frac{1}{2}M_4^{\uparrow 25} + \frac{1}{2}\pi_{\mathrm{proto}}\big)$, dilates it into a smooth envelope

$$\mathcal{E} = \mathrm{AvgPool}_{3\times3}\big(\mathrm{MaxPool}_{5\times5}(P_{\mathrm{sem}})\big), \tag{5}$$

and uses both maps to modulate the decoder output:

$$\begin{aligned}Z = \Big(Z_{\mathrm{dec}} &- \lambda_{\mathrm{out}}\,\mathrm{ReLU}\big(t_{\mathrm{out}} - \mathcal{E}^{\uparrow 88}\big)\\ &+ \lambda_{\mathrm{in}}\,\mathrm{ReLU}\big(P_{\mathrm{sem}}^{\uparrow 88} - t_{\mathrm{in}}\big)\Big)^{\uparrow 352}.\end{aligned} \tag{6}$$

The first term lowers logits outside the envelope, and the second raises them in confident lesion cores.

TABLE 1. Cohorts after deduplication. Tiny: lesion under 1.5% of the frame; camouflaged: under 3% with low colour contrast across the lesion boundary. Adverse corruptions: NBI-like spectral shift; low light (γ, vignetting, sensor noise); specular glare with blur; and the three combined.

Cohort	Source	Images	Role
Train	Kvasir-SEG [3]	800	training
40-label subset	Kvasir-SEG	40	training
Validation	Kvasir-SEG	95	selection
Kvasir test	Kvasir-SEG	90	in domain
PolypGen-Dev	PolypGen [36]	258	external
PolypGen-Ext	PolypGen	974	held out
PolypGen-Ext-SD	PolypGen sequence-disjoint subset	900	held out
PraNet-Ext	CVC-300, ClinicDB, ColonDB, ETIS [10]	696	external
ETIS	ETIS-LaribPolypDB	196	external
Tiny/camouflaged	PolypGen-Dev, ETIS, ColonDB (188 + 122)	310	stress
Adverse	PolypGen-Dev, 4 corruptions	258	stress

F. TRAINING OBJECTIVE

DANG and our in-house controls are trained with

$$\begin{aligned}\mathcal{L} = \mathcal{L}^{w}_{\mathrm{BCE+Dice}}(Z, Y) + 0.25\,\mathcal{L}^{w}_{\mathrm{Tv}}(Z, Y) + 0.1\,\mathcal{L}_{\mathrm{far}}(Z, Y)\\ + \tfrac{0.25}{3}\textstyle\sum_{j=2}^{4}\mathcal{L}^{w}_{\mathrm{BCE+Dice}}(M_j^{\uparrow 352}, Y) + 0.35\,\mathcal{L}_{\mathrm{proto}}.\end{aligned} \tag{7}$$

Each image's terms are weighted by w, which grows as lesion area shrinks. The Tversky term $\mathcal{L}_{\mathrm{Tv}}$ penalises false negatives more for small lesions. $\mathcal{L}_{\mathrm{far}}$ is the mean predicted probability outside dilations of the ground truth. The three coarser decoder logits $M_2, \ldots, M_4$ receive deep supervision. $\mathcal{L}_{\mathrm{proto}}$ applies the cross-entropy, Dice and Tversky terms to π_{proto} against the mask average-pooled to 25×25 and is present only in models with the prototype head. Exact forms are in Appendix A. The retrained published models use their official losses; one Polyp-PVT is additionally trained with Eq. (7) as a loss-matched control.

IV. EXPERIMENTAL SETUP

A. DATA

We remove from every evaluation set the images that duplicate training images, using MD5 hashes and a 64-bit perceptual difference hash (dHash $\leq$ 4); this includes Kvasir frames that public composites redistribute under other names. Table 1 lists the cohorts. "Tiny/camouflaged" and "adverse" are diagnostic slices of the external cohorts, not independent patient cohorts, and the adverse cohort is synthetic. We write "$40 \rightarrow X$" for a model trained on the 40-label subset and evaluated on cohort X, and "$800 \rightarrow X$" for the full training set. The 40-label subset is a fixed 5% random draw of the 800 training images, identical for all seeds and models.

PolypGen-Ext was not inspected during design and serves as a confirmation cohort. Because 74 of its frames share video sequences with PolypGen-Dev, we also report its sequence-disjoint subset PolypGen-Ext-SD. ClinicDB is unseen because training uses Kvasir only.

B. TRAINING AND MODELS

All models are trained at 352×352 with batch size 16, AdamW, warm-up followed by cosine decay, gradient clipping, mixed precision, random scaling, flips, 90° rotations and mild colour jitter, for 40 epochs on the 40-label subset and 20 epochs on the 800 images. After each epoch we evaluate both the raw weights and their exponential moving average on the validation set and keep the better checkpoint; test sets are never used for selection. Appendix A lists all hyperparameters.

PraNet, Polyp-PVT and EMCAD use their official architectures, ImageNet-pretrained backbones (Res2Net-50 and PVTv2-B2) and official losses, with our optimiser, schedule, augmentation and evaluation. Our in-house models are a ConvNeXt-Tiny + FPN baseline; a *concatenation control* that adds the same frozen DINOv2 and adapters as DANG and fuses the two streams by concatenation into an FPN decoder; DANG without the stem and style calibration; and DANG.

C. METRICS AND STATISTICS

Predictions are thresholded at 0.5. mDice is the mean of per-image Dice scores. Boundary F1 (BF1) [37] matches predicted and ground-truth contours within 2 pixels. Appendix C reports the 95th-percentile Hausdorff distance (HD95).

Five seeds ($n = 5$) are the unit of analysis. For a given seed, all models see the same subset and data order, so differences are tested with two-sided paired t-tests. For every pair of models we correct the 12 primary comparisons (mDice, BF1 and HD95 on the three 40-label external cohorts; mDice on 800 $\rightarrow$ETIS, PraNet-Ext and PolypGen-Ext) with Holm's procedure [38]. Figure intervals are 95% t-intervals. Inference speed is measured at batch size 1 in FP16 on an NVIDIA T4.

V. RESULTS

A. COMPARISON WITH RETRAINED PUBLISHED MODELS

Table 2 compares DANG with PraNet, Polyp-PVT and EMCAD retrained under our protocol.

a: 40 labels.

DANG has the highest mDice and BF1 of all retrained models on all three external cohorts. Its margin over the best retrained model in each column is 7.1–7.6 pp mDice and 4.8–10.0 pp BF1; the best competitor is PraNet in four columns and a Polyp-PVT variant in two. Against the models trained with their official losses, every BF1 difference and every adverse-cohort mDice difference survives Holm correction. The tiny/camouflaged and PraNet-Ext mDice differences to Polyp-PVT do not (adjusted $p = 0.085$ for each). The Polyp-PVT trained with our loss does not close the gap in mean, but its seed-to-seed SD reaches 12.7 pp and none of its differences to DANG is significant.

b: 800 labels.

DANG still beats PraNet on every cohort. Its differences to Polyp-PVT ($+1.4$ pp on ETIS, $+0.7$ pp on PraNet-Ext, -0.1 pp on PolypGen-Ext) and to EMCAD are not significant. DANG's advantage is therefore specific to the label-scarce regime.

c: Qualitative results.

Figure 3 shows predicted masks on randomly drawn frames. Every model misses the tiny ColonDB lesion. On the camouflaged, low-light and combined-corruption frames DANG reaches Dice 82%, 53% and 60%, and on the low-light frame it is the only model that finds the lesion. On the NBI frame DANG (63%) is below the baseline (67%) and PraNet (65%).

B. STANDARD TRAINING SPLIT

Our protocol trains on Kvasir only, so its numbers are not comparable with the literature, which trains on 900 Kvasir and 550 ClinicDB images. To link the two, we trained DANG and Polyp-PVT on this standard split for 50 epochs, used the final moving-average weights without any checkpoint selection, and evaluated at native resolution with three seeds (Table 3). Our Polyp-PVT reproduction is within 1.1 pp of its published scores on every test set, which validates the pipeline. DANG matches it on Kvasir, ClinicDB, ColonDB and CVC-300 (differences within ± 1.1 pp, not significant) and exceeds it on ETIS by 5.5 pp mDice ($p = 0.026$) and 11.1 pp BF1 ($p = 0.007$). These two tests are paired over three seeds and uncorrected, and ETIS was among the cohorts inspected during design (Section VI); the result is consistent with, but weaker evidence than, the five-seed results.

C. WHERE THE GAIN COMES FROM

The in-house rows of Table 2 isolate the components; Appendix B gives the paired differences (Table 5) and per-seed plots (Figures 4 and 5).

a: Frozen features alone do not explain the gain.

The concatenation control has the same frozen DINOv2 and adapters as DANG and 120M parameters, yet with 40 labels it stays at the level of the 31M-parameter baseline (44.9% vs. 44.7% on tiny/camouflaged, 46.3% vs. 45.6% on adverse). DANG exceeds it by 6.1 pp ($p = 0.003$) and 9.5 pp ($p < 0.001$) mDice on these cohorts and by 14.1 pp BF1 on tiny/camouflaged lesions, all significant after correction. The gain therefore lies in the components that consume the features, the prototype head, PASSE and the strip decoder together with the prototype loss. Because the control also uses a simpler decoder, this comparison does not isolate any single component.

b: Gains over the baseline.

With 40 labels, DANG raises BF1 by 11.9–14.4 pp on all three external cohorts and mDice on the adverse cohort by 10.2 pp, all significant after correction; the 6.3 pp tiny/camouflaged mDice gain is not ($p = 0.014$, adjusted 0.055). With 800 labels DANG gains 6.3 pp on ETIS and 3.2 pp on PraNet-Ext, and on the adverse cohort mDice

TABLE 2. With 40 labels DANG leads every retrained model; with 800 labels it ties the strongest. mDice and BF1 (%), mean ± SD over 5 seeds; "40 →" columns train on 40 Kvasir images, "800 →" columns on 800. Bold: best in column. *: worse than DANG after Holm correction (Section IV-C). Params: total parameters in millions (DANG: 43.3M trained, 86.6M frozen). FPS: batch size 1, FP16, NVIDIA T4.

			40 → Tiny/camouflaged		40 → Adverse		40 → PraNet-Ext		800 →, mDice		
Model	Params	FPS	mDice	BF1	mDice	BF1	mDice	BF1	ETIS	PraNet-Ext	PolypGen-Ext
Published models, retrained under our protocol											
PraNet [10]	30.5	40.6	$40.4_{\pm 1.5}$*	$21.6_{\pm 1.5}$*	$48.5_{\pm 3.5}$*	$22.4_{\pm 1.6}$*	$62.9_{\pm 1.7}$*	$32.2_{\pm 1.2}$*	$69.7_{\pm 2.0}$*	$76.7_{\pm 1.1}$*	$76.9_{\pm 0.9}$*
Polyp-PVT [11]	25.1	42.4	$42.9_{\pm 4.0}$	$14.8_{\pm 4.2}$*	$39.3_{\pm 4.5}$*	$10.7_{\pm 2.6}$*	$64.3_{\pm 4.5}$	$21.5_{\pm 6.5}$*	$77.5_{\pm 2.2}$	$81.1_{\pm 0.8}$	$\mathbf{80.2}_{\pm 1.6}$
Polyp-PVT, loss of Eq. (7)	25.1	43.0	$43.9_{\pm 10.2}$	$16.4_{\pm 6.8}$	$38.3_{\pm 11.4}$	$12.1_{\pm 4.9}$	$62.2_{\pm 12.7}$	$22.7_{\pm 9.2}$	$77.2_{\pm 1.0}$	$80.4_{\pm 0.8}$	$79.4_{\pm 1.0}$
EMCAD [12]	26.8	36.5	$29.6_{\pm 3.1}$*	$6.0_{\pm 2.5}$*	$29.2_{\pm 3.5}$*	$3.3_{\pm 1.1}$*	$49.7_{\pm 4.9}$*	$8.0_{\pm 4.1}$*	$77.5_{\pm 2.1}$	$81.0_{\pm 0.9}$	$79.7_{\pm 0.9}$
In-house models											
ConvNeXt-T + FPN	31.0	91.9	$44.7_{\pm 3.8}$	$16.2_{\pm 3.9}$*	$45.6_{\pm 4.0}$*	$15.2_{\pm 2.0}$*	$66.6_{\pm 3.8}$	$28.4_{\pm 4.8}$*	$72.5_{\pm 2.7}$*	$78.7_{\pm 1.5}$*	$78.7_{\pm 0.8}$*
Concatenation control	120.2	38.4	$44.9_{\pm 0.8}$*	$16.6_{\pm 2.5}$*	$46.3_{\pm 1.8}$*	$16.4_{\pm 1.6}$*	$67.2_{\pm 1.2}$*	$29.2_{\pm 4.2}$*	$72.9_{\pm 3.9}$	$79.7_{\pm 1.0}$*	$79.3_{\pm 1.4}$
DANG without stem	129.4	19.3	$50.1_{\pm 1.4}$	$27.4_{\pm 1.6}$	$\mathbf{57.2}_{\pm 2.0}$	$25.6_{\pm 1.1}$	$71.5_{\pm 0.8}$	$40.4_{\pm 1.0}$	$77.5_{\pm 0.7}$	$\mathbf{81.9}_{\pm 0.8}$	$79.7_{\pm 0.7}$
DANG	129.9	17.4	$\mathbf{51.0}_{\pm 1.7}$	$\mathbf{30.7}_{\pm 1.2}$	$55.8_{\pm 1.2}$	$\mathbf{27.2}_{\pm 1.7}$	$\mathbf{71.9}_{\pm 0.3}$	$\mathbf{42.2}_{\pm 1.6}$	$\mathbf{78.9}_{\pm 1.5}$	$81.8_{\pm 0.6}$	$80.1_{\pm 0.4}$

TABLE 3. Standard training split (1,450 images). mDice (%) at native resolution and ETIS BF1; mean ± SD over 3 seeds for our runs. Published Polyp-PVT scores from [11]. Bold: better of our two runs.

	mDice					BF1
Model	Kvasir	ClinicDB	ColonDB	ETIS	CVC-300	ETIS
Polyp-PVT (published)	91.7	93.7	80.8	78.7	90.0	–
Polyp-PVT (ours)	$\mathbf{92.2}_{\pm 0.6}$	$92.6_{\pm 0.3}$	$\mathbf{80.6}_{\pm 0.7}$	$78.9_{\pm 1.2}$	$89.9_{\pm 0.2}$	$56.8_{\pm 1.6}$
DANG	$91.9_{\pm 0.3}$	$\mathbf{93.7}_{\pm 0.7}$	$80.2_{\pm 0.9}$	$\mathbf{84.4}_{\pm 0.4}$	$\mathbf{90.8}_{\pm 1.1}$	$\mathbf{67.9}_{\pm 0.1}$

rises from 55.8% for the baseline to 67.4% (Figure 4). In-domain accuracy is unchanged (92.95% vs. 92.92% on Kvasir test). On the held-out PolypGen-Ext the gain is smaller but significant (+1.4 pp, $p = 0.009$; +1.2 pp micro-averaged on PolypGen-Ext-SD, $p = 0.004$).

c: The stem shows no measurable gain.

Removing the Retinex stem and style calibration keeps almost all of the improvement, and no difference between DANG and its stem-free variant survives correction. Uncorrected, the stem costs 4.3 pp mDice on NBI-shifted frames ($p = 0.008$) and adds 3.3 pp BF1 on tiny/camouflaged lesions ($p = 0.022$). A plausible reason for the NBI loss, which we did not test, is that the stem standardises colour towards white-light Kvasir statistics. We report the full model as DANG because all main experiments use it, but recommend the stem-free variant, which is also faster.

D. EFFICIENCY

DANG is the most expensive model in Table 2: 129.9M parameters, of which 86.6M are frozen DINOv2 weights, and 17.4 FPS against 36.5–42.4 FPS for the retrained models. The concatenation control runs the same two encoders at 38.4 FPS, so most of DANG's extra latency lies in its decoding path rather than in the foundation model. At 17–19 FPS DANG suits second-reader and offline review. Real-time use at 25–30 Hz would need a faster GPU, an optimised decoding path, or distillation into a smaller network [23], [24].

VI. LIMITATIONS

1) **Training length of the retrained models.** With 40 labels the retrained models reached their best validation score in the last 5 of 40 epochs and may not have converged; DANG peaked between epochs 25 and 39. Longer training could narrow the gap.
2) **One subset, one source centre.** The 40-image subset is the same for all seeds, so the reported variance reflects optimisation, not the choice of images; all main models are trained on Kvasir only.
3) **Adaptive design.** PASSE thresholds and loss constants were tuned while inspecting PolypGen-Dev, ETIS and the two stress cohorts, so these are not fully held out; this includes ETIS, the cohort of the standard-split result. PolypGen-Ext was not used for design, and gains there are smaller.
4) **Shared sequences.** 74 PolypGen-Ext frames share video sequences with PolypGen-Dev; PolypGen-Ext-SD removes them.
5) **Synthetic and derived cohorts.** The adverse cohort is synthetic rather than paired physical NBI/white-light acquisitions, and both stress cohorts are slices of the external cohorts.
6) **Statistical power.** Five seeds give four degrees of freedom and the standard-split experiment three; several raw $p < 0.05$ effects do not survive correction.
7) **Confounded prototype loss.** Models with the prototype head also receive $\mathcal{L}_{\text{proto}}$, and we have not run a factorial over PASSE, the prototype loss and the strip decoder.
8) **Scope.** We evaluate single frames, not video, and report no clinical reader study.
9) **Cost.** DANG is 2–2.5× slower than the retrained models and has 4–5× more parameters.

VII. CONCLUSION

DANG uses a frozen DINOv2 through a prototype margin head, a prototype-anchored support envelope and an uncertainty-guided strip decoder. Trained on 40 labelled images it outperforms PraNet, Polyp-PVT and EMCAD re-

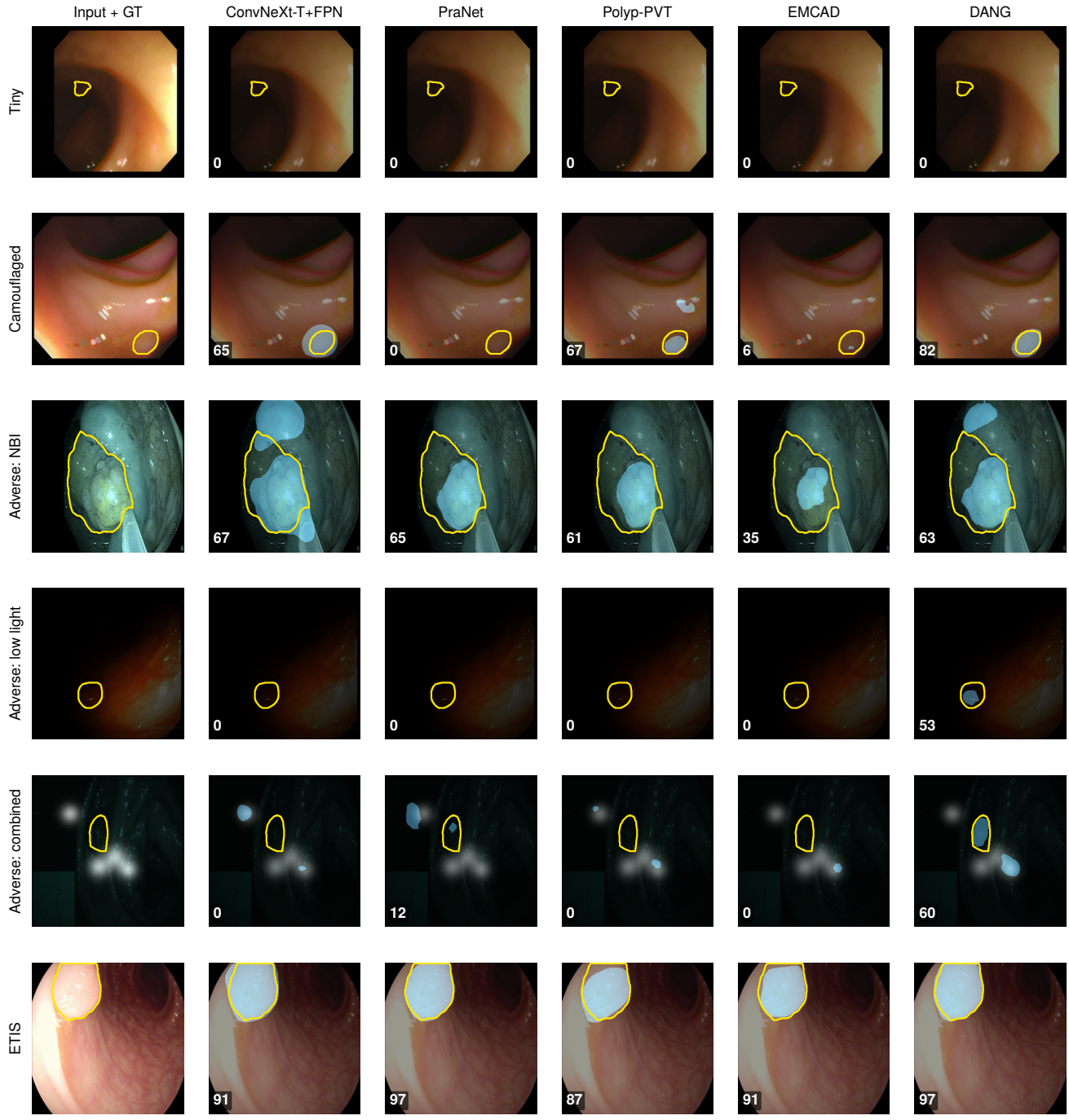


FIGURE 3. Predicted masks with 40 training labels. Each row is one frame drawn at random (not curated) from the named cohort; columns are the ConvNeXt-T baseline, PraNet, Polyp-PVT, EMCAD and DANG, all trained on the same 40 Kvasir images with the same seed. Blue fill: prediction at threshold 0.5; yellow contour: ground truth; number: per-image Dice (%).

trained under the same protocol on every external cohort, by at least 7.1 pp mDice and 4.8 pp BF1 over the best of them, and on the standard split it matches Polyp-PVT on four test sets and exceeds it on ETIS. A same-size concatenation control shows that the frozen features alone do not explain the gain. With 800 labels DANG ties the best retrained models, and it is 2–2.5× slower. Next steps are longer training of the baselines in the label-scarce regime, resampled subsets, a lighter backbone and decoding path, and evaluation on physical NBI acquisitions and video.

APPENDIX A IMPLEMENTATION DETAILS

APPENDIX B ABLATION DETAILS

TABLE 4. Hyperparameters. Lesion area a is the ground-truth foreground percentage of the image.

Item	Setting
Backbones	ConvNeXt-Tiny (ImageNet-22k, fine-tuned on ImageNet-1k at 384); DINOv2-Base (LVD-142M), frozen
Adapters	convolutional LoRA, rank 4, on DINOv2 blocks 8–11
Optimiser	AdamW; learning rate 1.6×10^{-4} (CNN), 4×10^{-4} (new modules), 10^{-3} (adapters)
Weight decay	10^{-4}; none on normalisation, bias, adapter and prototype parameters
Schedule	linear warm-up (3 epochs with 40 labels, 2 with 800), then cosine decay to 5%
Epochs, batch	40 (40 labels), 20 (800 labels), 50 (standard split); batch size 16
Weight averaging	EMA with decay $\min(0.992,\ (1+t)/(10+t))$ at step t
Other	gradient-norm clipping at 3.0; FP16 on T4, BF16 on newer GPUs
Augmentation	scale 0.65–0.90 ($p = 0.3$) or 0.95–1.20 with crop or pad; horizontal and vertical flips ($p = 0.5$ each); 90° rotations ($p = 0.5$); brightness, contrast, saturation ±15% ($p = 0.4$); hue ±0.04 ($p = 0.3$); gamma 0.8–1.25 ($p = 0.3$)
Image weight	$w = \mathrm{clip}\big((3/(a+0.35))^{0.4},\ 1,\ 2.4\big)$, normalised to mean 1 in the batch
Tversky	false-positive weight $0.45 + 0.15\,\sigma((a-1.5)/0.5)$, false-negative weight 1 minus it
Far-field loss	mean probability outside the ground truth dilated by 4, 8, 14 and 20 pixels
Prototype head	$K = 8+8$ prototypes, dimension 128; $\beta = 6$; τ learned, initialised at 12 and clamped to $[4, 25]$
	$w_{\mathrm{a}} = 10 \cdot \frac{1}{2}\tanh(\theta_{\mathrm{a}})$, $w_{\mathrm{ss}} = 8 \cdot \frac{1}{2}\tanh(\theta_{\mathrm{ss}})$ with learned θ
Anchors	momentum 0.95; foreground where pooled mask ≥ 0.35, background where ≤ 0.10
PASSE	$t_{\mathrm{out}} = 0.22$, $\lambda_{\mathrm{out}} = 10$, $t_{\mathrm{in}} = 0.38$, $\lambda_{\mathrm{in}} = 1.2$
Adverse cohort	NBI: fixed linear colour mixing that attenuates red; low light: gamma 2.55, gain 0.45, radial vignette, Gaussian noise (SD 0.035); glare and blur: Gaussian blur (radius 3.2), contrast ×0.55, four specular spots; combined: all three with milder low light (gamma 2.15) and blur (radius 2.2)
Camouflaged lesion	area $< 3\%$ and colour distance < 0.085 between 3-pixel bands inside and outside the boundary

TABLE 5. Paired differences between in-house models, five seeds; p-value of a two-sided paired t-test in parentheses. Bold: significant after Holm correction over the 12 columns of the row. B: ConvNeXt-T + FPN; S: concatenation control; C: DANG without stem; F: DANG. HD95 in pixels at 352 × 352 (lower is better).

	40 → Tiny/camouflaged			40 → Adverse			40 → PraNet-Ext			800 →, mDice		
	mDice	BF1	HD95	mDice	BF1	HD95	mDice	BF1	HD95	ETIS	PraNet-Ext	PolypGen-Ext
F−B	+6.3 (.014)	**+14.4** (.002)	−8.1 (.135)	**+10.2** (.005)	**+11.9** (.002)	**−8.4** (.002)	+5.3 (.038)	**+13.8** (.007)	−9.2 (.258)	**+6.3** (.005)	**+3.2** (.003)	**+1.4** (.009)
F−S	**+6.1** (.003)	**+14.1** (<.001)	−10.9 (.017)	**+9.5** (<.001)	**+10.8** (.001)	**−9.5** (.004)	**+4.6** (.002)	**+13.0** (.004)	−13.7 (.040)	+5.9 (.016)	**+2.1** (.002)	+0.7 (.347)
C−B	+5.3 (.017)	**+11.1** (.001)	−7.0 (.187)	**+11.7** (.003)	**+10.4** (<.001)	**−15.8** (<.001)	+5.0 (.037)	**+11.9** (.003)	−9.4 (.207)	+4.9 (.019)	+3.3 (.019)	+1.0 (.127)
F−C	+1.0 (.218)	+3.3 (.022)	−1.1 (.551)	−1.4 (.112)	+1.5 (.225)	+7.3 (.012)	+0.3 (.489)	+1.9 (.152)	+0.2 (.896)	+1.4 (.106)	−0.1 (.847)	+0.4 (.471)

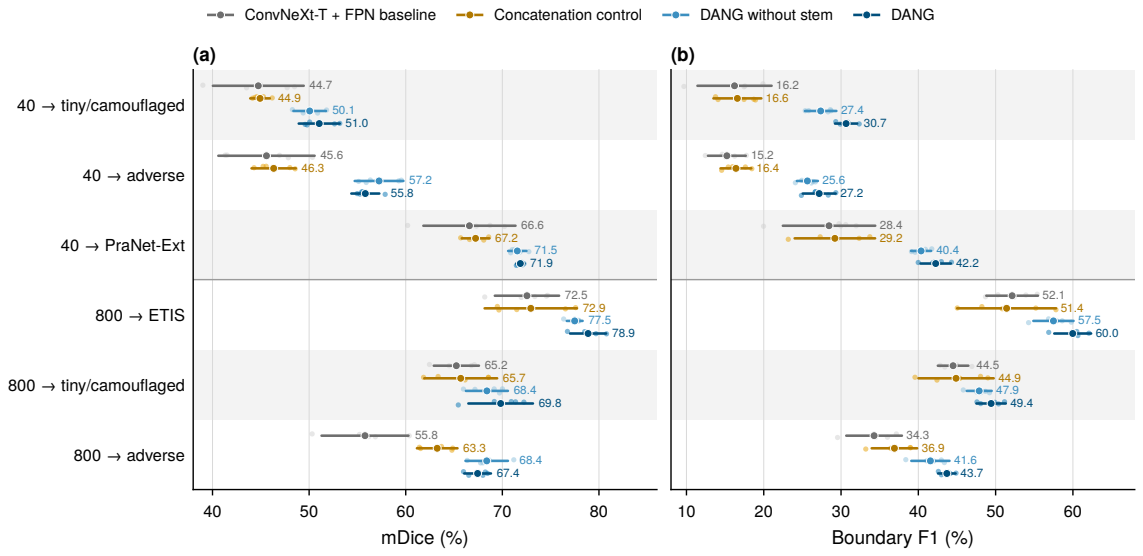


FIGURE 4. In-house models per seed. (a) mDice and (b) BF1. Large dots and printed values are means over 5 seeds, horizontal lines are 95% t-intervals of the mean, faint dots are individual seeds. The concatenation control (orange) stays at baseline level with 40 labels, while both models with the prototype head and PASSE (blue) separate from it.

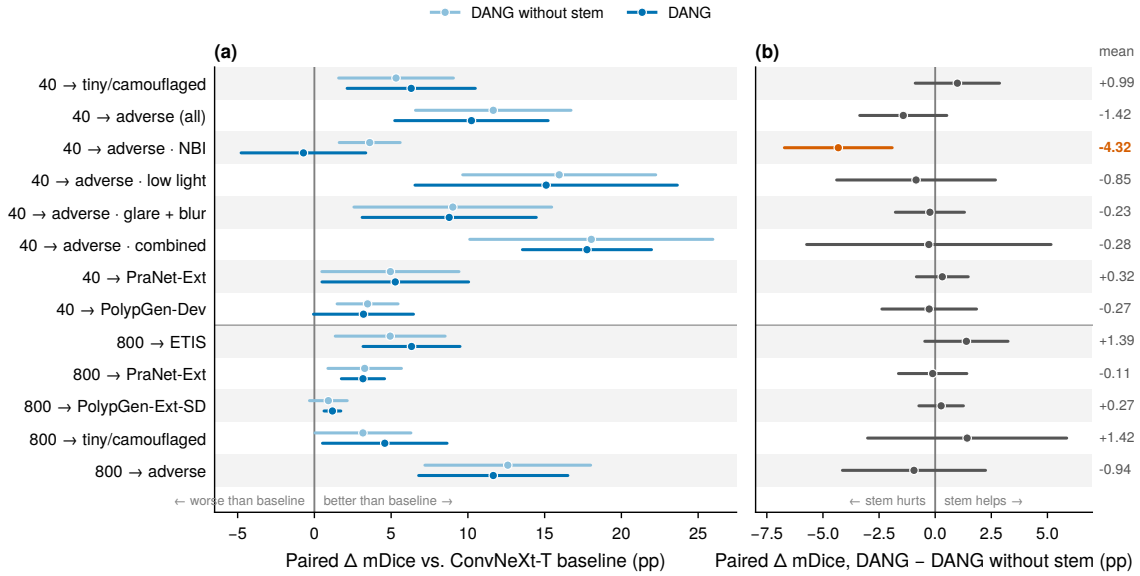


FIGURE 5. Contribution of the Retinex stem and style calibration. (a) Paired mDice difference to the baseline for DANG with and without the stem. (b) Paired difference between the two. Points are means over 5 paired seeds; bars are 95% *t*-intervals. In (b) the one interval excluding zero is a 4.3 pp loss on NBI frames ($p = 0.008$); no difference in (b) survives Holm correction over its 13 rows.

APPENDIX C HD95 AND EMPTY PREDICTIONS

When a prediction is empty and the ground truth is not, HD95 is undefined; we assign a quarter of the image diagonal, $0.25 \cdot 352\sqrt{2} \approx 124.5$ px. This penalty is smaller than the HD95 of 10.9% of non-empty predictions, so an empty mask can score better than a mask with a distant false positive. Empty-prediction rates are similar for DANG and the baseline (28.5% vs. 27.8% on $40 \rightarrow$tiny/camouflaged). With half- or full-diagonal penalties the direction of the HD95 differences holds but their size shrinks, and on two cohorts they are not significant even with the default penalty (Table 5). We therefore base our claims on mDice and BF1.

DATA AND CODE AVAILABILITY

All experiments use public, de-identified datasets (Kvasir-SEG [3], PolypGen [36], CVC-ClinicDB, CVC-ColonDB [7], CVC-300 and ETIS-LaribPolypDB [6]) under their research licences; no new patient data were collected. Code, deduplication manifests, per-image evaluation logs and the scripts that regenerate every table and figure from the raw logs will be released.

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